FORT

Three Phase Multifunction Network Power Meter

User Manual

E12ZY (D)

Contents

1.Product brief introduction ……1

2.Function introduction ……2

3.Technical parameter ……3

4.Installation and Wiring ……4

5.Programme operation ……8

6.Panel explain and measurement information display ……12

7.Communication Protocol ……16

8.Function Output ……29

9.Ordinary problem and solution ……33

1. Product Brief introduction

1.1 Standard

Reference standard

GB/T 17883-1999 Class 0.2S and 0.5s Static AC Active Electric Meter

GB/T 17882-1999 Class 2 and 3 Static AC Reactive Electric Meter

DL/T 614 Multifunction Electric Power Meter

GB/T 13850 Ac Energy Transfer to Analog Quantity or Electric Measuring Transmitter of Digital Signal.

Performance Standard:

GB/T 22264.1- Panel Type Digital Display Electric Measuring Instrument, Part 1: Definition and General Requirement.

GB/T 22264.2- Panel Type Digital Display Electric Measuring Instrument, Part 2: Special Requirements for Current Meter and Voltage meter.

GB/T 22264.3- Panel Type Digital Display Electric Measuring Instrument, Part 3: Special requirements for power meter and reactive power meter.

GB/T 22264.4- Panel Type Digital Display Electric Measuring Instrument, Part 4: Special requirements for frequency meter.

GB/T 22264.5- Panel Type Digital Display Electric Measuring Instrument, Part 5: Special requirement for phase meter and power factor meter

GB/T 22264.7- Panel Type Digital Display Electric Measuring Instrument, Part 7: Define and General requirement.

GB/T 22264.8-Panel Type Digital Display Electric Measuring Instrument, Part 8: Commend test way.

1.2 Product Overview

Three phase multifunction network power meter which provided with high ability and precision can measure all electrical parameters in three phase electric network, include three phase voltage, three phase current, active power, reactive power, apparent power, frequency, power factor, four-quadrant energy, UIPQ demand quantity, time-sharing accounting, switching input monitoring, electric pulse output and with communication interface.

The product is provided with many extended function of input and output for choosing:1 communication interface, 3 oops of transmitting output, local or remote switch signal output,3 loops of switch monitoring, 2 loops electric pulse output.

The product is with best competitive, can directly replace with general power transmitted, measuring meter, power energy measure meter and relative auxiliary unit, can widely apply to many fields, like energy manage system, power monitor system, industry and mining enterprise, public place, intelligent building and switch board those power net system, SGADA system and so on. The product is provided with many advantages such as simple installation, simple wiring, convenient maintenance, less engineering work, programming on the spot and it can complete different communication software network for PLC, industry control.

2. Function Introduction (See Table 1)

Table 1

Measurement function		Remark
Real-time measurement	Three phase voltage	Basic function
	Three phase current	
	Power, Frequency, Power factor	
Energy measurement	Active energy	
	Reactive energy	
	Apparent energy	
	Two-way measurement	
Communication	RS485 interface MODBUS-RTU	
Switching Input	Passive dry contact point	Extended function need to be customized extraly
Switching Output	AC250V 5A remote/alarm	
Display mode		LCD display

Remarks: Attention please! This user manual is a common used User Manual,it mentions all the customized extended functions,such as Relay Output etc, but only a few of the above mentioned extended functions might be made in the real product,for the function doesn't has in the real product, it means that functions are not customized yet.

3. Technical parameter (See Table 2)

Table 2

Item			Parameter
Signal input	Wiring		3 phase 4 lines/3 phase 3 lines
	Voltage	Measurement range	450VAC(L-N) / 750VAC(L-L)
		Over load	Continuous: 1.2 multiple, instantaneous: 2 multiple/1s
		Power consumption	<1VA (per phase)
	Current	Measurement range	AC(50mA~5A)
		Over load	Continuous: 1.2 multiple, instantaneous: 10 multiple/5s
		Power consumption	<0.4VA (per phase)
	Frequency		45~65Hz
Power supply			200~240 Vac , ≤5VA
Communication			RS485 communication interface, physical layer isolation. Meet the international standard of MODBUS-RTU agreement Communication speed 1200~38400 Test type N81,E81,081
Switching output			Programmable remote/alarm switching output Capacity: 5A/250VAC 5A/30VDC Programmable alarm power or remote mode
Remote switch			Remote switch input measuring, passive dry contact input, programmable relate alarm output
Measurement class			Power: 0. 5 Frequency: ±0. 2Hz Active power: 1. 0 Reactive power: 2. 0
Display mode			LCD display
Environment			Working temperature: -10~+45°C Storage temperature: -25~+50°C Relative humidity: <85%RH
Safety			Isolation:resistance of Signal input, power source and output terminal to cover>100MΩ Withstand Voltage:Signal Input/power supply , Power Supply/Signal output :AC 2kV, Signal Input/Signal Output:1kV
Protection grade			Front panel: IP52

4. Installation and Wiring

4.1 Panel Size(mm): 96×96 Hole Size(mm): 91×91 Depth(mm): 60 (See Fig.1)

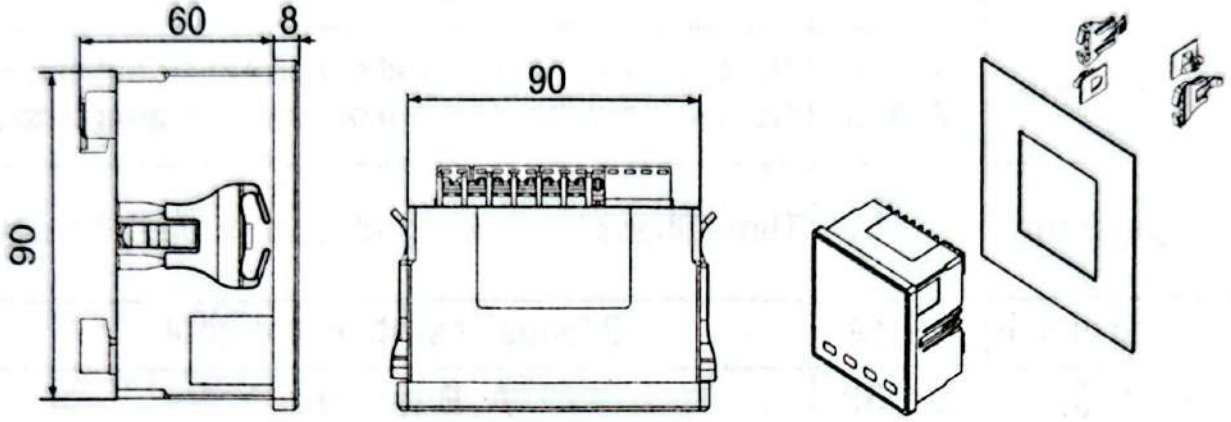


Fig.1

4.2 Panel Size(mm): 80×80 Hole Size(mm): 76×76 Depth(mm): 71 (See Fig.2)

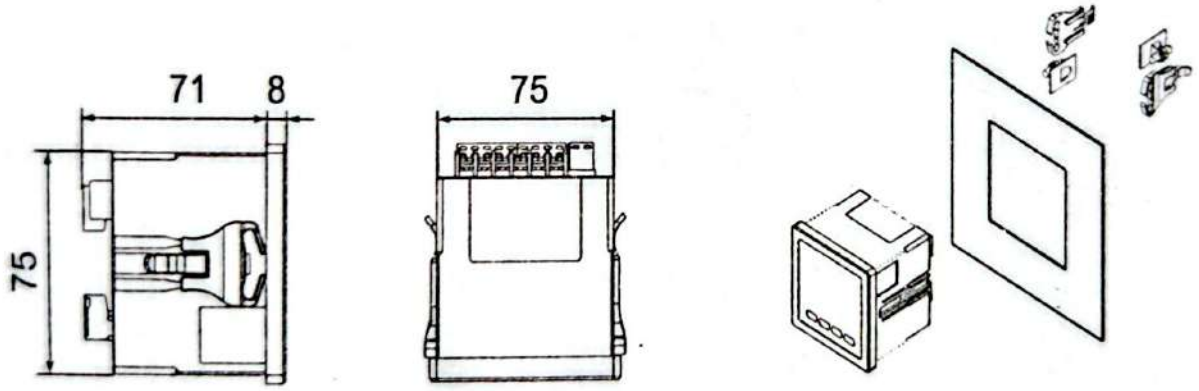


Fig.2

4.3 Panel Size(mm): 72×72 Hole Size(mm): 67×67 Depth(mm): 71 (See Fig.3)

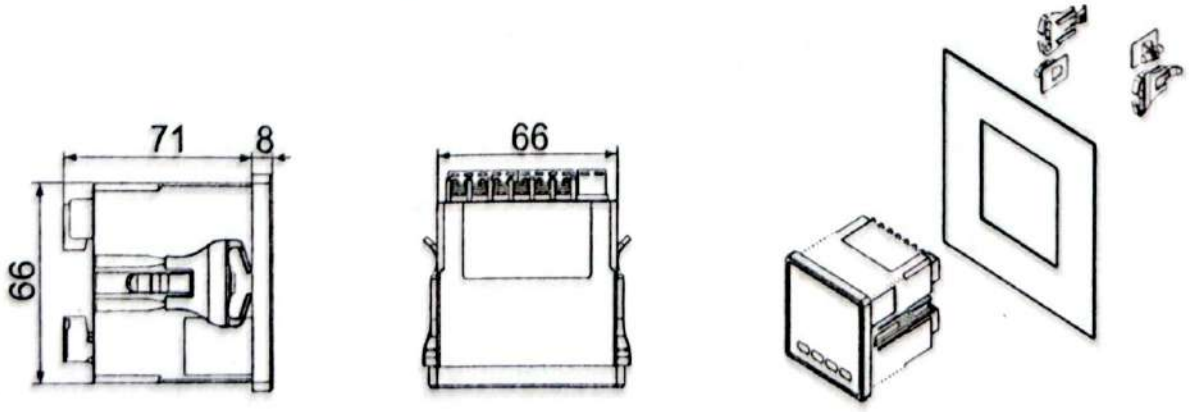


Fig.3

4.4 Wiring terminal function (see Table 3)

Table 3

Power supply	1, 2	200~240 Vac
Current signal	4, 5, 6 7, 8, 9	No.4,6,8 are the terminal of current signal input No.5,7,9 are the terminal of current signal output
Voltage signal	11, 12 13, 14	Three phase Voltage signal input is Ua, Ub,Uc,Un
Switching output	15-18	2 loops of switching output
RS485	58, 59	A, B terminal
Switching input	70-74	4 loops of switching input, 70 is the common terminal

Instructions for use

(a) 1, 2 is the auxiliary power supply terminal for meter working, please make sure the supply power is suitable for this product to avoid damaging product.

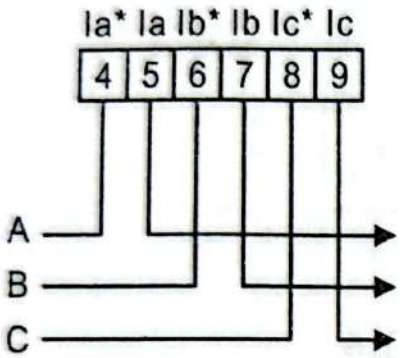
(b) 4,6,8 is the input terminal of current transformer, with* signal is for the input terminal of current.

(c)Three phase three lines connection: Current of B phase no need connection, UB connect to NO.14 terminal. Refer to 4.4 wiring.

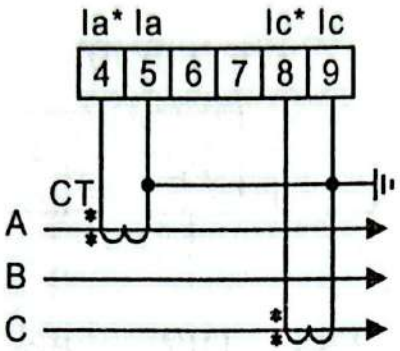
(d) Product connection refer to the wiring diagram printed on the product cover.

4.5 Wiring (see Fig.4 and FIg.5)

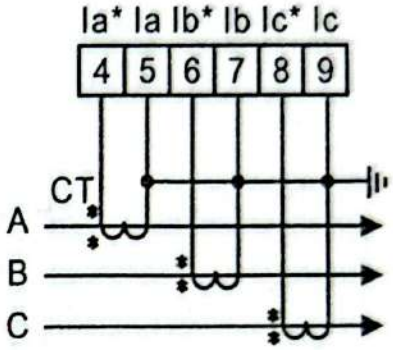
Fig.4



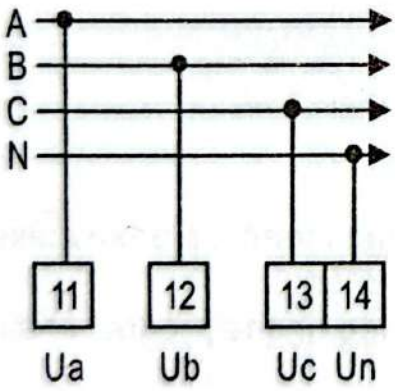
3 phase AC(without CT)



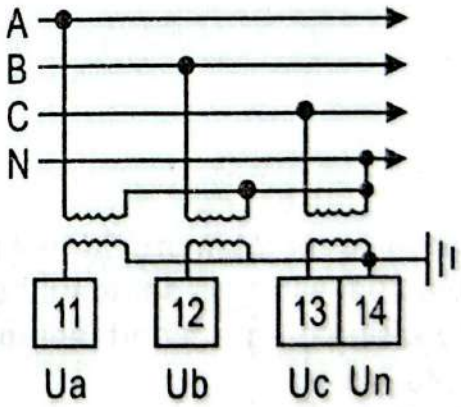
3 phase AC(2CT)



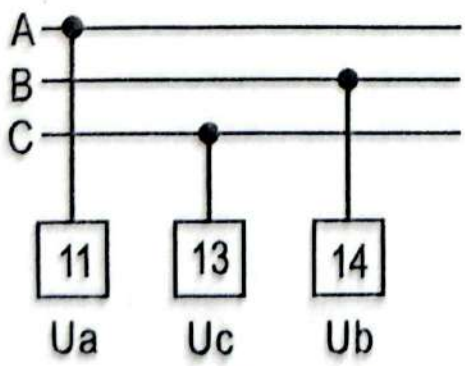
3 phase AC(3CT)



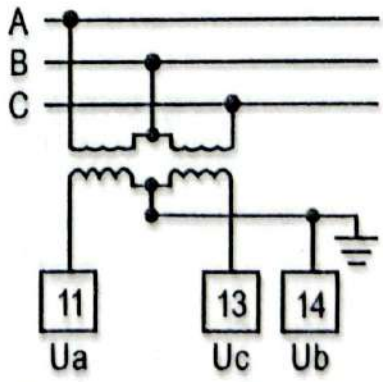
Low voltage:3 phase 4 wire AC(without PT)



High voltage:3 phase 4 wire AC(3PT)



Low voltage:3 phase 3 wire AC(without PT)



High voltage:3 phase 3 wire AC(2PT)

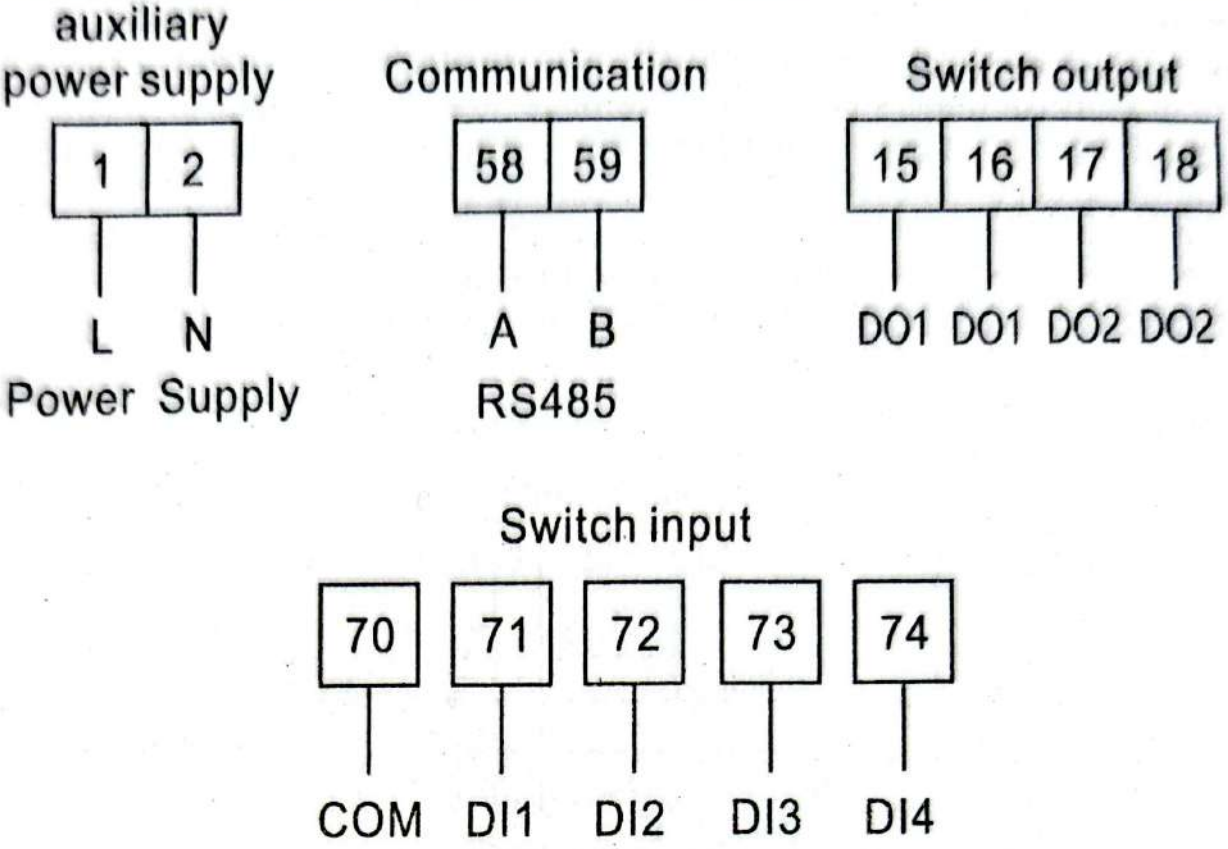


Fig.5

Note: The meter is provided with two kinds of wiring modes, please make sure the actual wiring is same to the set wiring in the meter.
The wiring diagram and technical parameter printed on the product shall be prevailed.

5. Programme operation

Under programming status,
No.1 line is first layer menu information.
No.2 line is second layer menu information,
No.3 line is third layer menu information.

For example: as shown in Fig.1,
No.1 line: INPT means signal input.
No.2 line:CT current ratio
No. 3 line:80 means the CT value

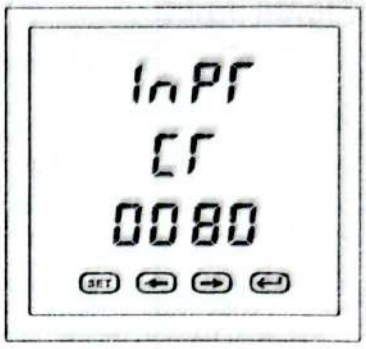


Fig.6

CT value=CT Primary value/ Secondary value
Example,if the Current Transformer is 400/5A,
The CT value=400/5=80,
we need to input 0080 as CT ratio.

Parameter setting can be customized according to requirements.(See Table 4)

Table 4

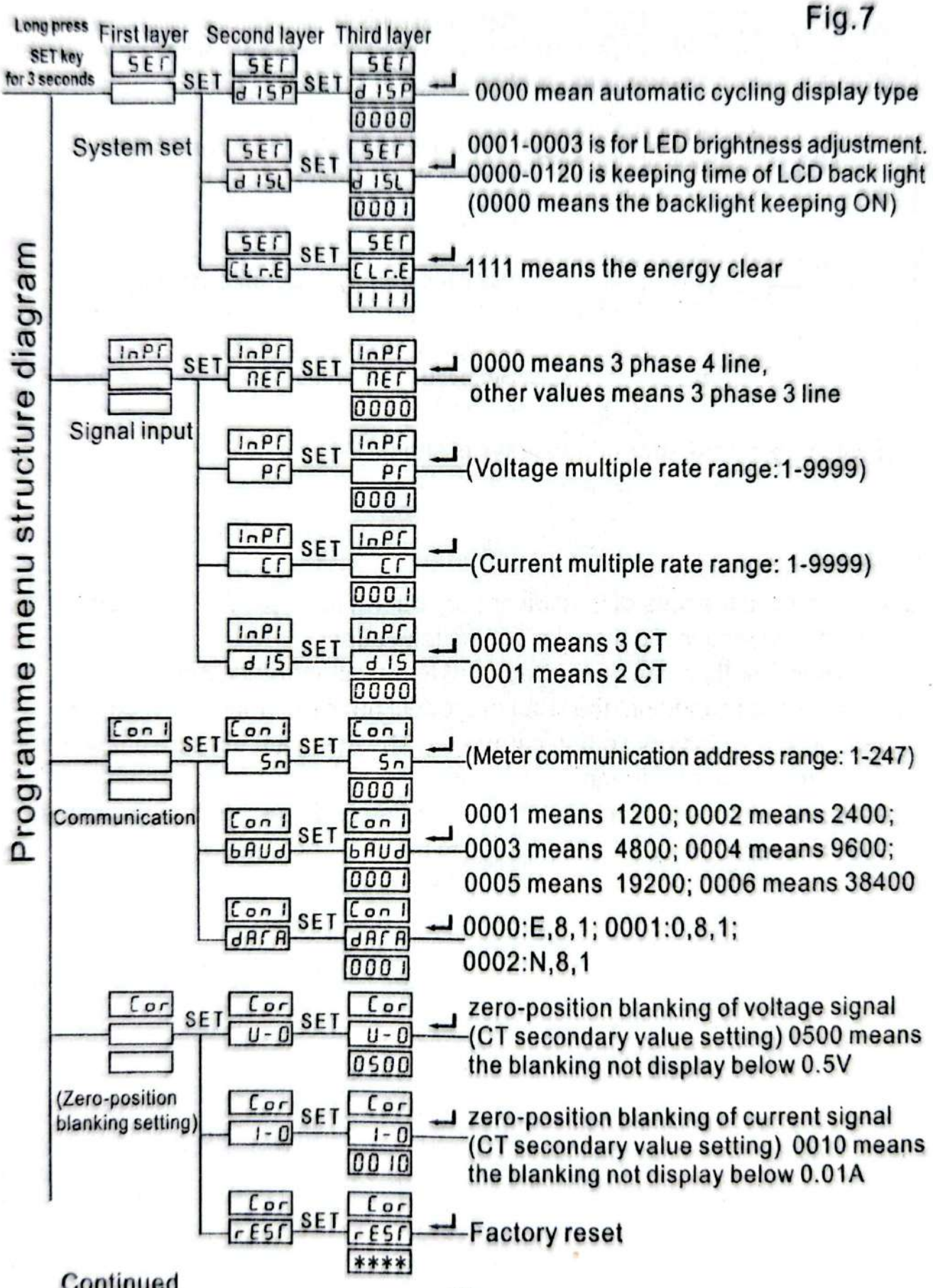
No.1 layer	No.2 layer	No.3 layer	Description
System set "SET"	Display DISP	0000-0014	0000 means automatic cycling display(Displayed information see Table 5,6)
	DISL	0001-0003 or 0000-0120	0001-0003 is for LED brightness adjustment. 0000-0120 is keeping time of LCD back light(0000 means the backlight keeping ON)
	Energy clear CLr.E	1111	1111 means the energy clear, other values are invalid

Signal input INPT	Wiring mode NET	0000 or other values	0000 means 3 phase 4 line, other values means 3 phase 3 line
	Voltage transformation ratio PT	1~9999	PT value= PT primary value/ secondary value
	Current transformation ratio CT	1~9999	CT value= CT primary value/ secondary value
Commun -ication set CONi (i is 1~2)	address SN	1~247	Meter address range: 1-247
	Communication speed BAUD rate	0001~0006	0001 means 1200; 0002 means 2400; 0003 means 4800; 0004 means 9600; 0005 means 19200; 0006 means 38400
	Data format DATA	0000~0002	0000:E,8,1; 0001:0,8,1; 0002:N,8,1
Switching output set D0-i (i is 1-4)	Choose alarm item or close alarm (refer to 8.2 switching output)	Set the specific threshold value of alarm item	Choose alarm item,and set relative threshold value(when alarm item is switching value, noneed set threshold value), once meet the alarm conditions, switch output ON.

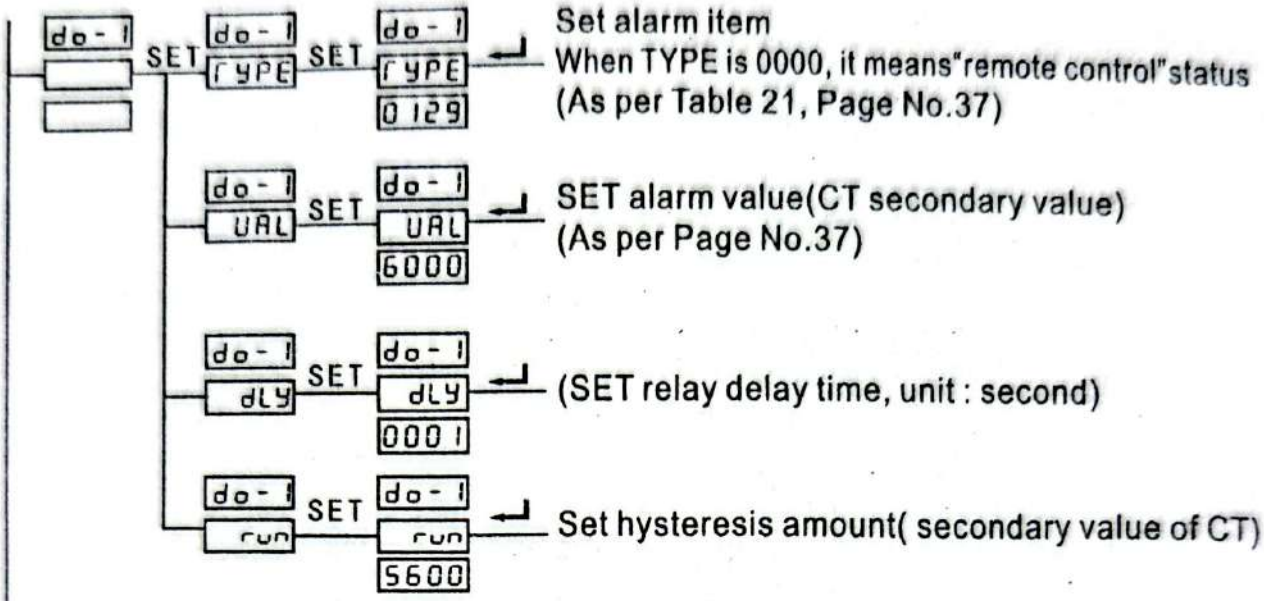
Note: The above menu is applied to the product with complete function.If you find there is no such menu in the product or the menu is not working, It means the product not supporting the function.

Programming step(see Fig.7)

Fig.7



To be continued



(Switching output: other three ways is similar)

Operation instruction:

(a) After revised the data of third layer (or option), need press " ↵ " button to back the second menu then the setting take effect.

(b)The wiring method can be revised refer to the actual wiring mode.

(c)Under normal condition, the label of product have remarked the model parameter and factory setting parameter. The user can reset the parameters according to the requirements.

(d) After revise the value , through " ← " button and " → " button to increase or decrease , through" **SET** " button to move.

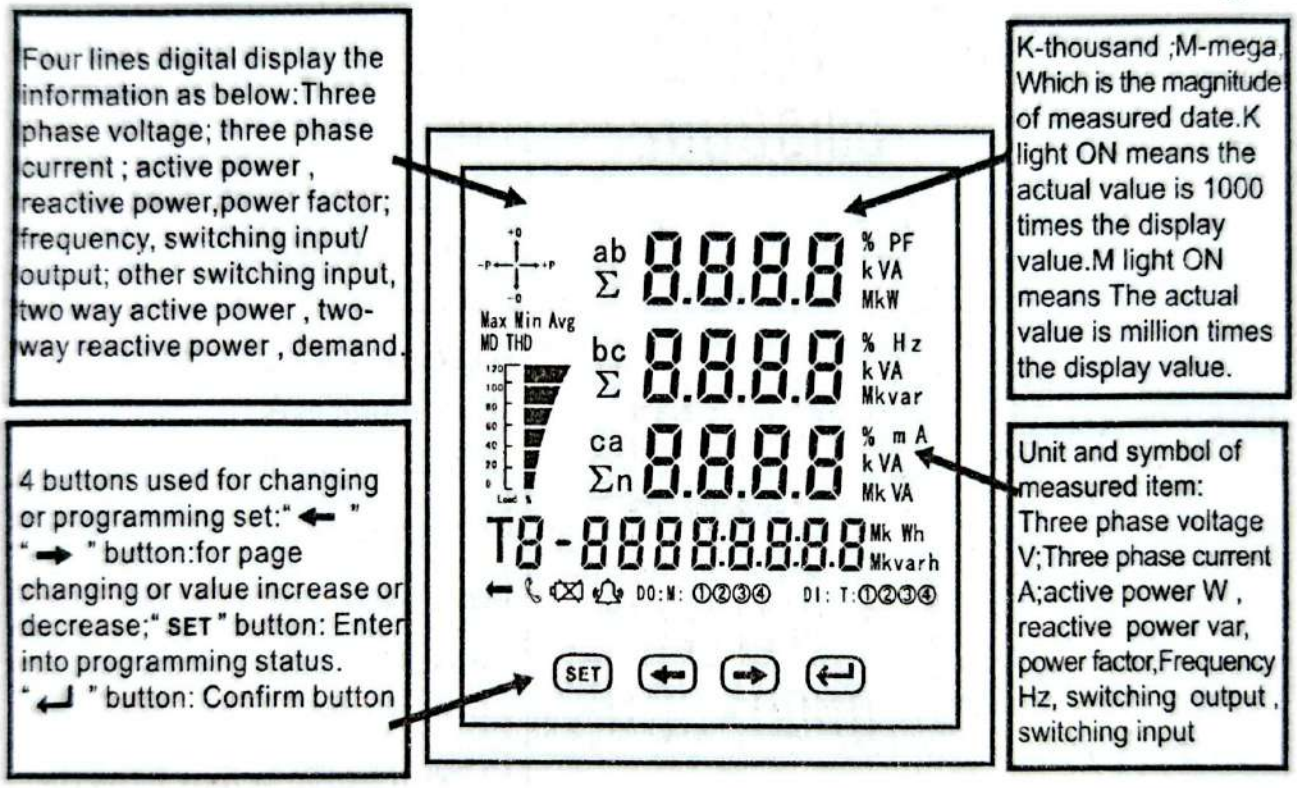
6. Panel explain and measurement information display

6.1 72×72,80×80,96×96 LCD display (see Fig.8)

(1) Panel instructions

If there is no relative symbol display (or the set data not working), It means the product is without the relative function.

Fig.8



(2) The contents of displayed page (see Table 5)

Table 5

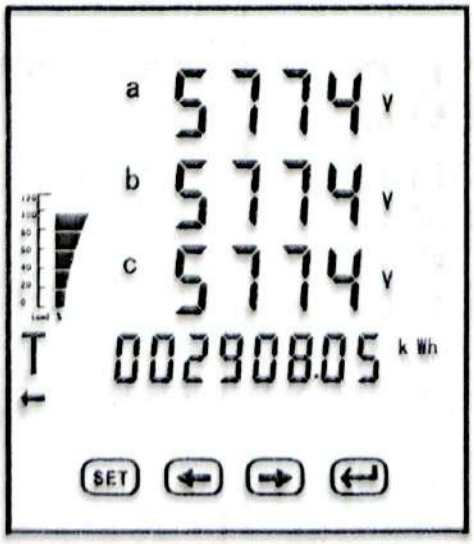
Page	Content	Explanation
DISP=1 Three phase voltage Positive Active Energy		Separately display the voltage Ua, Ub, Uc (3 phase 4 line), Positive Active Energy. In left Fig Ua=5774V Ub=5774V Uc=5774V Positive Active Energy =2908,05kWh

Table 5

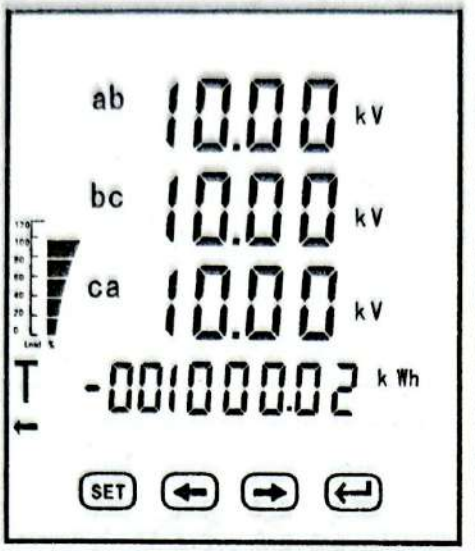
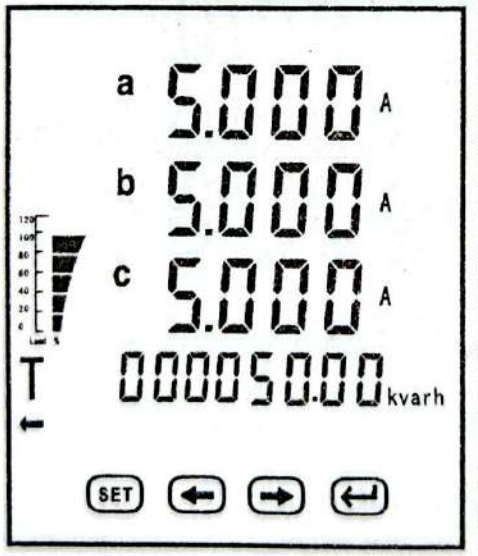
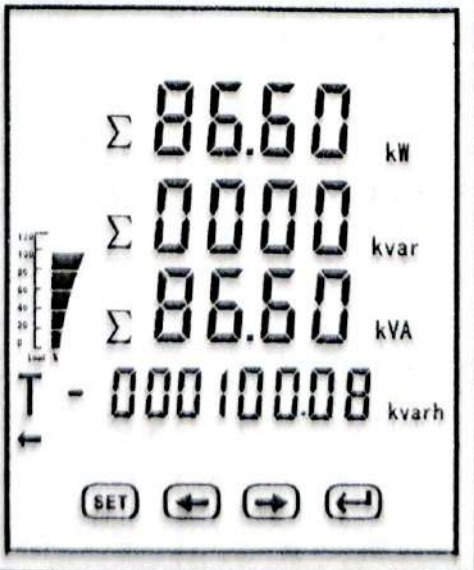
DISP=2 Three phase voltage Negative Active Energy		Separately display the voltage Uab, Ubc, Uca (line voltage), Negative Active Energy. In left Fig Uab=10kV Ubc=10kV Uca=10kV Negative Active Energy =1000.02kWh
DISP=3 Three phase current Positive Reactive Energy		Display 3-phase current Ia, Ib, Ic, the unit is A. Positive Reactive Energy In left Fig Ia=5A Ib=5A Ic=5A Positive Reactive Energy =50.00kvarh
DISP=4 Total active power Total reactive power Total apparent power Negative Reactive Energy		Total active power =86.60kW Total reactive power =0000kvar Total apparent power =86.60kVA Negative Reactive Energy =100.08kvarh

Table 5

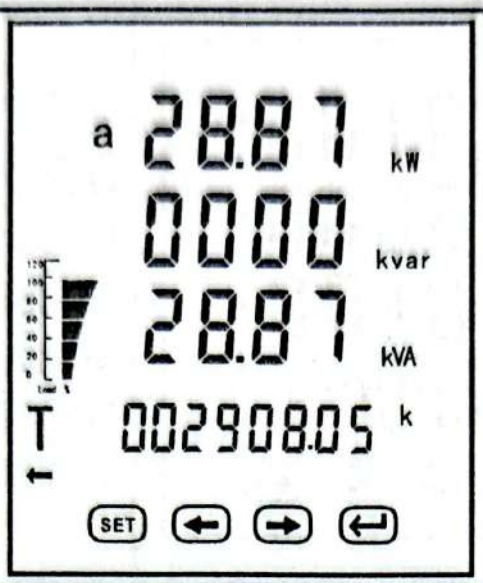
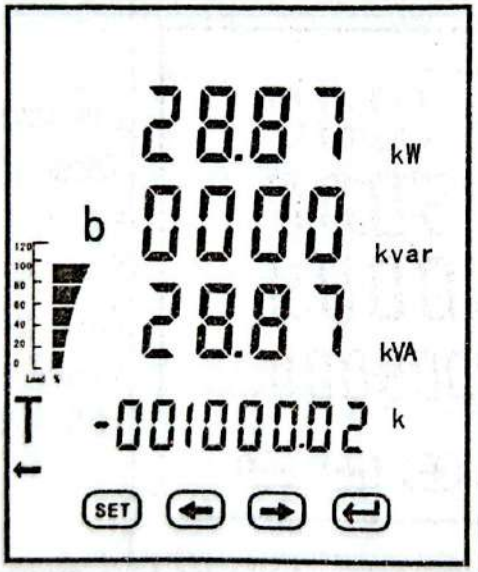
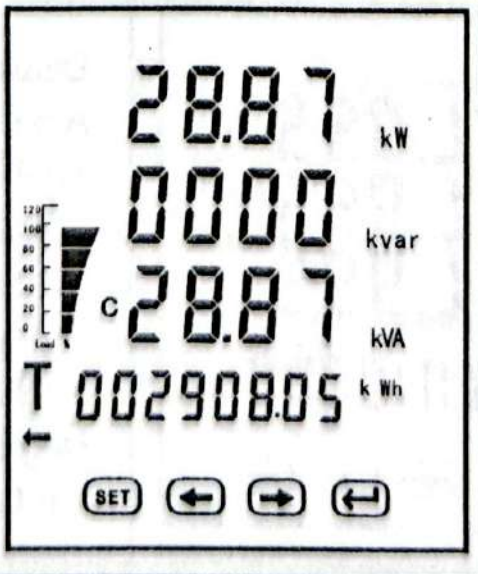
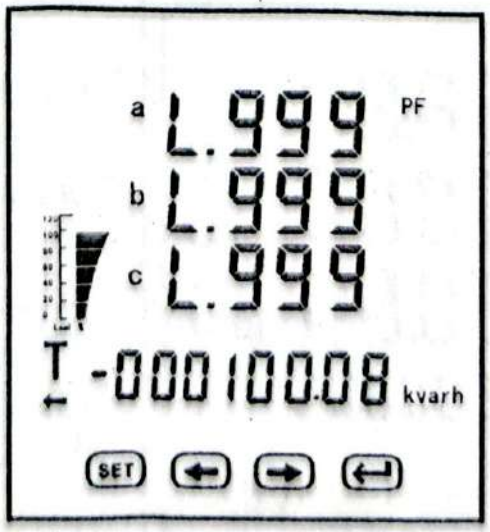
DISP=5 A phase active power A phase reactive power A phase apparent power Total Apparent Energy **("k" represents for kVAh)**		A phase active power =28.87kW A phase reactive power =0000kvar A phase apparent power =28.87kVA Total Apparent Energy("k") =2908.05kVAh
DISP=6 B phase active power B phase reactive power B phase apparent power Negative Apparent Energy **("k" represents for kVAh)**		B phase active power =28.87kW B phase reactive power =0000kvar B phase apparent power =28.87kVA Negative Apparent Energy("k") =1000.02kVAh
DISP=7 C phase active power C phase reactive power C phase apparent power Positive Active Energy		C phase active power =28.87kW C phase reactive power =0000kvar C phase apparent power =28.87kVA Positive Active Energy =2908,05kWh

Table 5

DISP=8 Current Unbalance Average current, Voltage Unbalance Negative Active Energy		Displayed in left Fig Current Unbalance=1% Average current=5A Voltage Unbalance=1% Negative Active Energy =1000.02kwh
DISP=9 Three phase total power factor Frequency, Average Voltage Positive Reactive Energy		Displayed in left Fig Three phase total power factor =1. 000 Frequency=50Hz Average Voltage=9V Positive Reactive Energy =50.00kvarh
DISP=10 Power factor for each phase Negative Reactive Energy		Displayed in left Fig A phase power factor =0.999 B phase power factor =0.999 C phase power factor =0.999 Negative Reactive Energy =100.08kvarh

7. Communication Protocol

7.1 physical layer

7.1.1 RS485 communication port, asynchronous half-duplex mode;

7.1.2 Communication speed 1200-38400bps can be set, factory default 9600dps;

7.1.3 Byte transfer format: 1 bit for initial bit, 8 bit for data bits, odd-even check (N81, E81, 081 can be selected), factory default N81.

7.2 Digital communication protocol

The meter is provided with serial asynchronism half-duplex RS485 communication port, adopt MODBUS-RTU protocol, various data can be transferred through communication line. One line can connect 32pcs meter at the same time, each meter can set different communication address. The communication terminal number of different series meters is different. communication should be connected by the shielded twisted-pair cable with copper network, and the diameter not less than 0.5mm^2.When wiring, keep the communication wire away from strong electric cables or other strong electric field, T type network Wiring is recommended (see Fig.10),Star-type or other wring is not recommended.

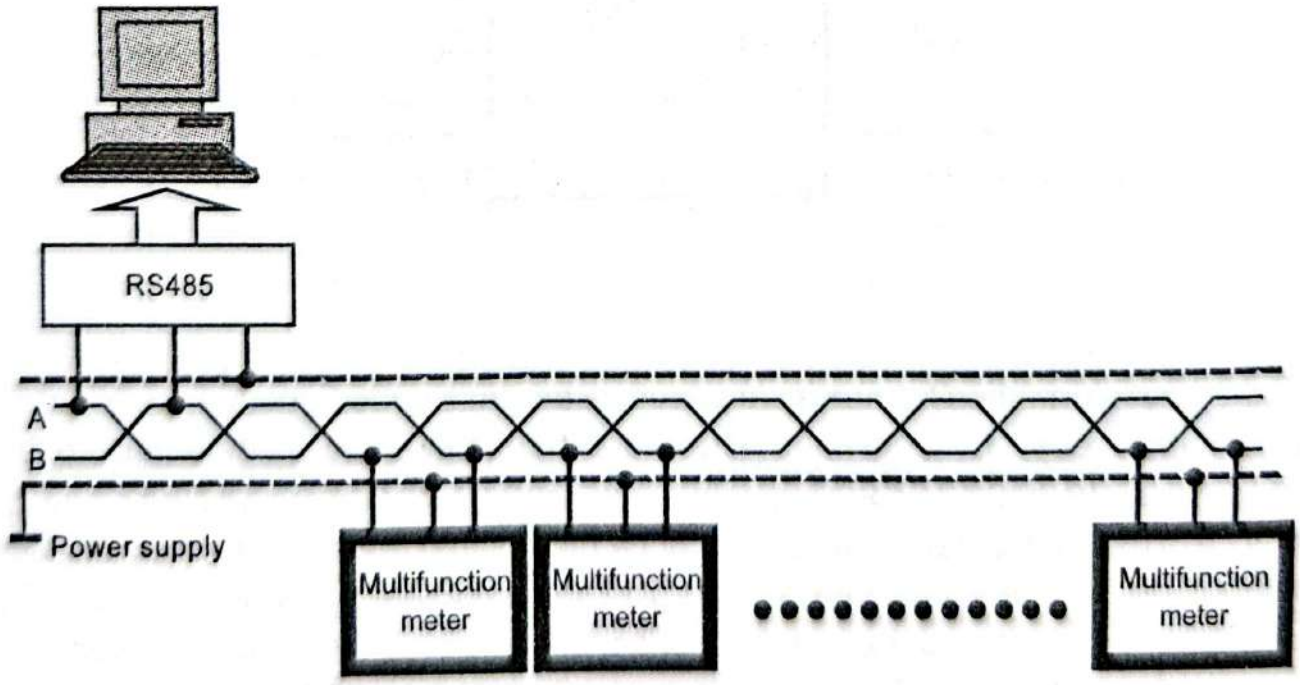
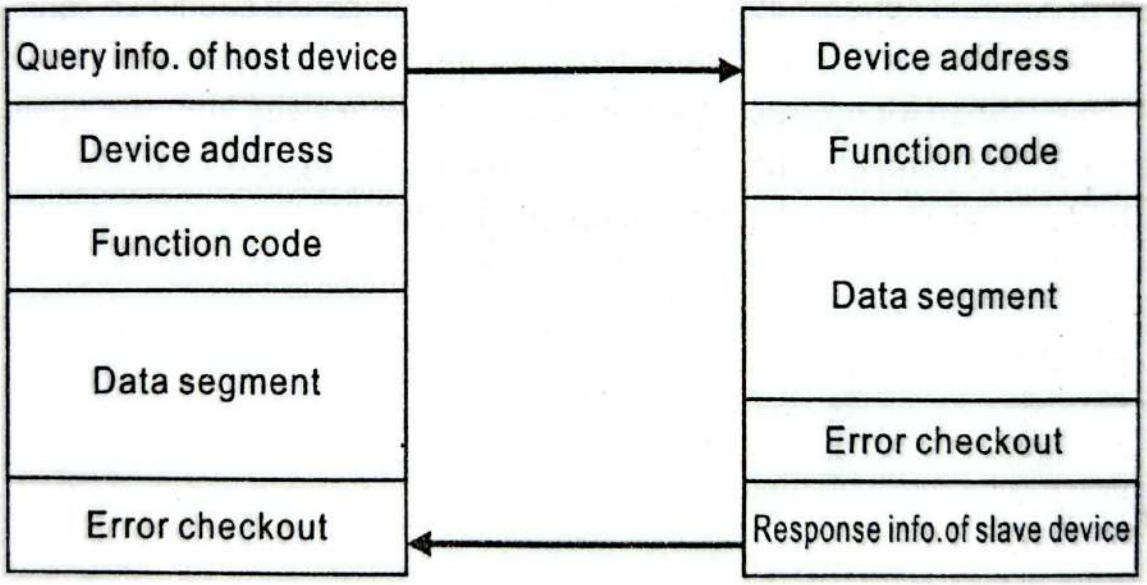


Fig.10

MODBUS-RTU communication protocol:

MODBUS protocol adopt host-slave response mode in one communication line. The signal of host computer addressing to the terminal device (Slave computer) with one address, then the terminal device send response signal back to host computer in the opposite direction. In one communication line, the communication data flow is transmitted in two opposite directions. (half-duplex working mode). MODUBUS protocol only allow the communication between Host computer (PC,PLC, etc) and terminal device(see Fig.11), not allow the data exchange between terminal devices, so each terminal device will not occupy the communication lines when initialized, but only response the query signal to the host computer.

Fig.11



Host PC enquiry:

The query information frames include device address code, function code, data information code ,checkout code. The address code indicate the selected slave PC device; The function code indicate the executed function of selected slave device, for example, function code 03 or 04 means that the slave device will read the register and send back the content; The data segment include the other additional information of executed function, for example, in reading command, reading from which register? How many register to be read? The checkout code used for checking the correctness of one frame information, providing the information verification method for slave device, it adopts CRC 16 checkout rules.

Slave PC response:
If slave PC respond normally, the information include the slave address code, function code, data code and CRC 16 checkout code. Data code include the collected data from slave device, for example, register value or status. If the error occurs, the slave device will not respond. The transmission mode means one series independence data structure in one data frame and the rules for data transmission. The below defines the transmission mode compatible with MODBUS protocol RTU mode. Each bit of byte: 1 start bit, 8 data bit,(odd-even check bit),1 stop bit (with odd-even check bit) or 1 stop bit (without odd-even check bit).

Data frame structure (Message format) see Table 7

Table 7

Address code	Function code	Data code	Checkout code
1 BYTE	1 BYTE	N BYTE	2 BYTE

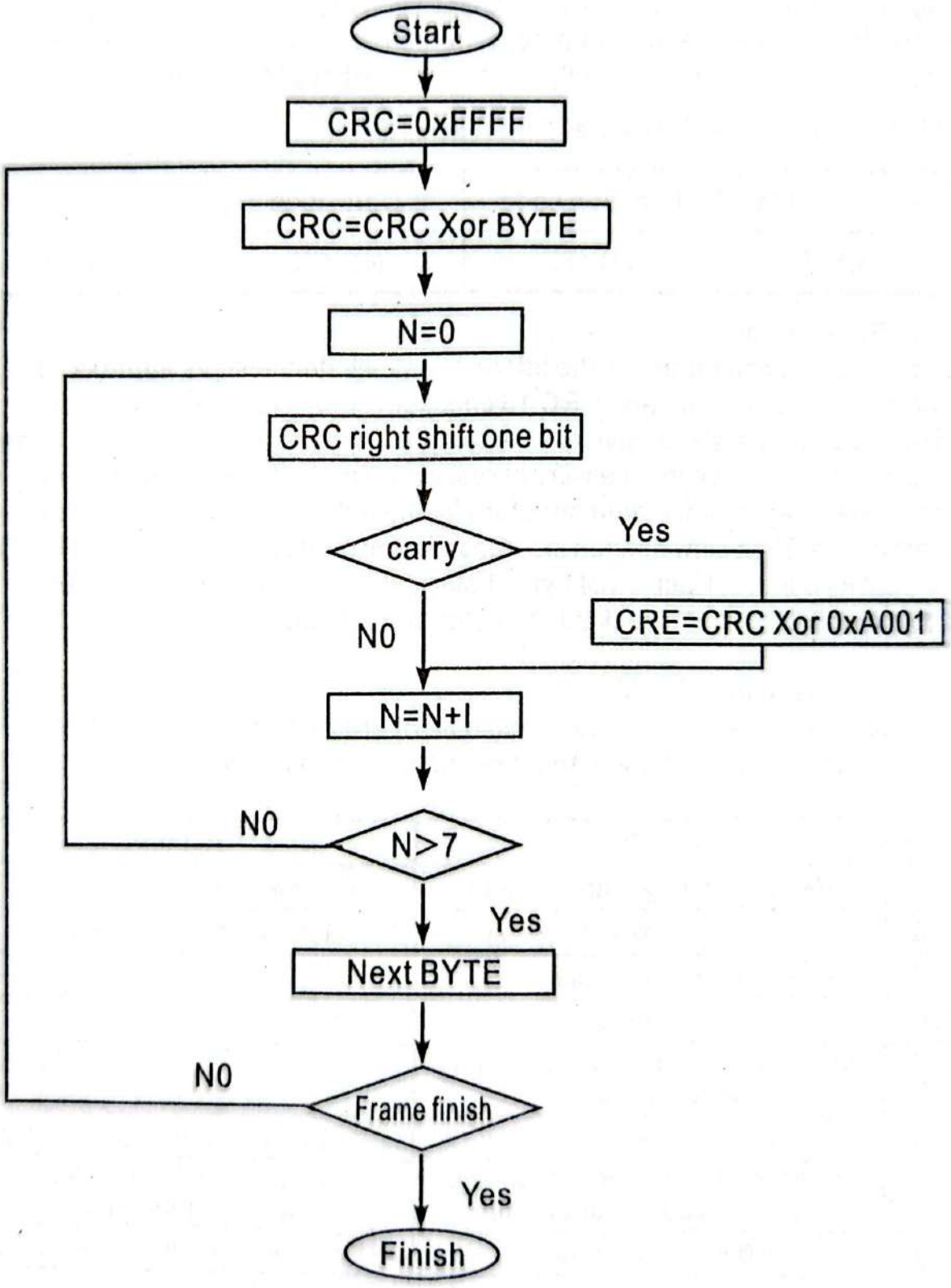
Slave PC response:
If slave PC respond normally, the information include the slave address code, function code, data code and CRC 16 checkout code. Data code include the collected data from slave device, for example, register value or status. If the error occurs, the slave device will not respond. The transmission mode means one series independence data structure in one data frame and the rules for data transmission. The below defines the transmission mode compatible with MODBUS protocol RTU mode. Each bit of byte: 1 start bit, 8 data bit,(odd-even check bit),1 stop bit (with odd-even check bit) or 1 stop bit (without odd-even check bit).

Function code: (see Table 8)
The code indicate which function should be operated by the terminal. Below Table shows the function code and their meaning and function.

Table 8

Code	Meaning	Action
01	Read switching output status	Get switching output status
02	Remote switch capacity input status	Get switching input status
03	Read the value of data register	
04	Read the value of data register	
05	Remotely control single switching output operation	
06	Write single register	Set the binary value to relative one register
0F	Remotely control more than one switching output operation	Get present binary value of one or more register
10H	Write preset register	Set the binary value to relative registers

Checkout code:
Error checkout (CRC) domain take 2 bytes, include one 16 bit binary value.CRC value is calculated by transmission device, then attached to data frame; When receiving device receive data, CRC value will be calculated again, then compare with the received value of CRC domain, if two values are not equal, the error occurs.



7.3 Message instruction format

Read switching output status 0 x 01 (see Table 9)

Table 9

Host PC query the command			Slave PC response	
Slave PC address	1 Byte	1 ~247	Slave PC address	1 Byte
Function code	1 Byte	0x01	Function code	1 Byte
Starting relay address	2 Bytes	0x0000 (Fixed)	Byte number of register	1 Byte
Relay quantity	2 Bytes	0x0004 (MAX)	Register value	N Bytes
CRC checkout code	2 Bytes		CRC checkout code	2 Bytes

Input status command of remote-check switch 0 x 02 (see Table 10)

Table 10

Host PC query the command			Slave PC response	
Slave PC address	1 Byte	1 ~247	Slave PC address	1 Byte
Function code	1 Byte	0x02	Function code	1 Byte
Starting switch address	2 Bytes	0x0000 (Fixed)	Byte number of register	1 Byte
Remote-check switch quantity	2 Bytes	0x000C (MAX.)	Register value	N Bytes
CRC checkout code	2 Bytes		CRC checkout code	2 Bytes

Command of remote-control single-loop switching output 0 x 05 (see Table 11)

Table 11

Host PC query the command			Slave PC response	
Slave PC address	1 Byte	1 ~247	Slave PC address	1 Byte
Function code	1 Byte	0x05	Function code	1 Byte
Starting relay address	2 Bytes	0x0000~ 0x0003	Starting relay address	2 Bytes
Relay action value	2 Bytes	0xFF00/ 0x0000	Relay action value	2 Bytes
CRC checkout code	2 Bytes		CRC checkout code	2 Bytes

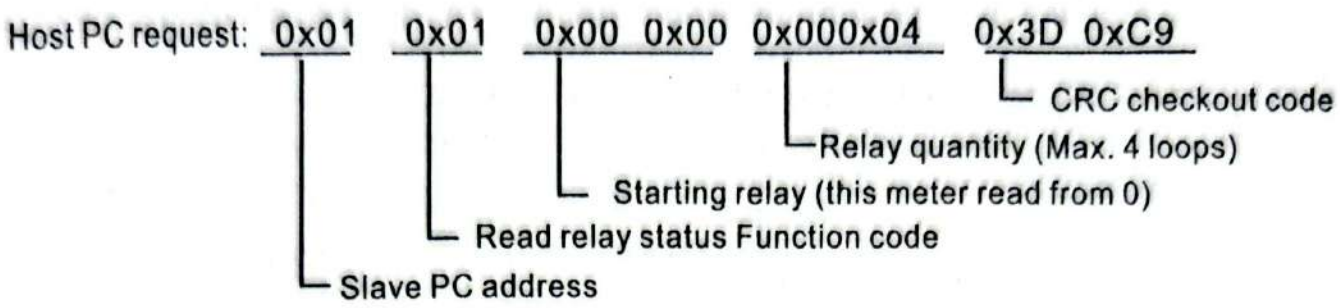
Output command of remote-check multi-loop switch 0 x 0F (see Table 12)

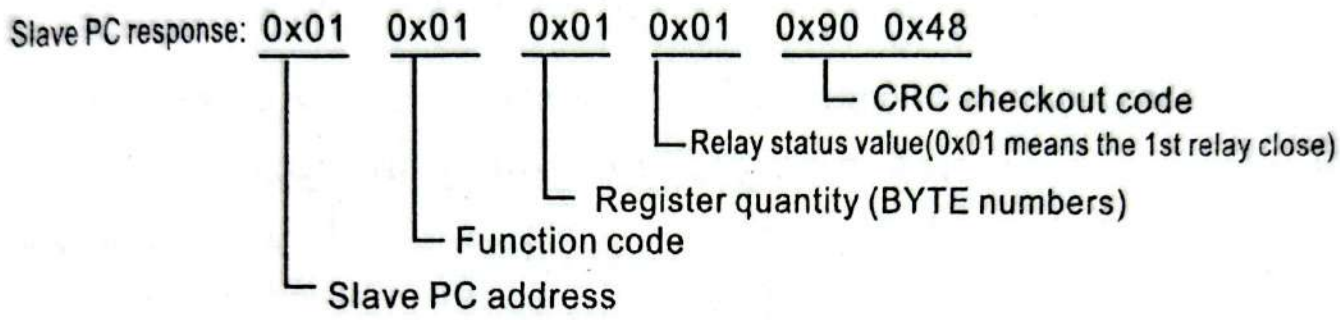
Table 12

Host PC query the command			Slave PC response	
Slave PC address	1 Byte	1 ~247	Slave PC address	1 Byte
Function code	1 Byte	0x0F	Function code	1 Byte
Starting relay address	2 Bytes	0x0000(Fixed)	Starting relay address	2 Bytes
Relay quantity	2 Bytes	0x0004(Fixed)	Relay quantity	2 Bytes
Byte numbers of data	1 Byte	0x01	CRC checkout code	2 Bytes
Multi-Relay action value	1 Byte			
CRC checkout code	2 Bytes			

7.4 Message examples

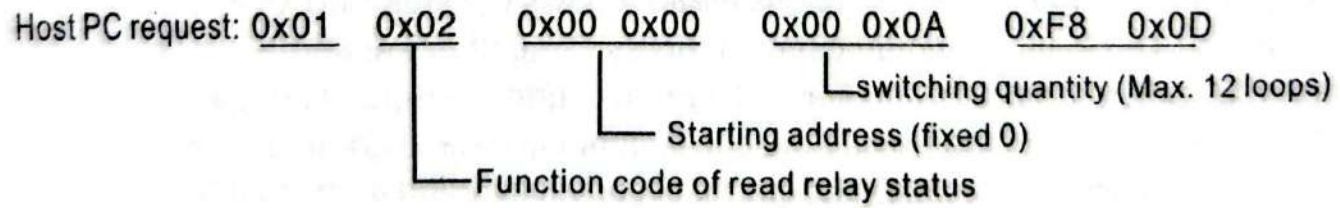
(1) Read remote-control/Alarm switching output status (Function code0 x 01)

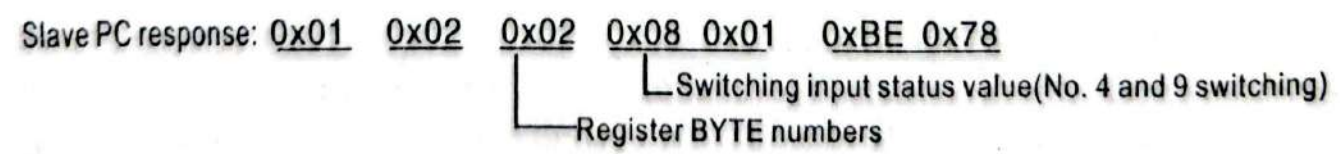




Description: The relay status value corresponds to the status value of each switch output starting from the lowest bit of each byte according to the MODBUS protocol. 1 means the ON status and 0 means the OFF status. Binary " 0000 0011" of "0x03" Indicates that the first and second relays are closed.

(2) Remote-check switching input status (Function code0 x 02)





Description: The switching input status value corresponds to the status value of each switching input starting from the lowest bit of each byte according to the MODBUS protocol. 1 means the ON status and 0 means the OFF status.

(3) Remote-control single switching output status (Function code 0 x 05)

NOTE:This meter has a maximum of 4 switching output. The use of remote control commands must require the relay to work in remote control mode.

Host PC request: 0x01 0x05 0x00 0x00 0xFF 0x00 CRC

- 0xFF 0x00 — Relay action value (0xFF00:OFF 0x0000:ON)
- 0x00 0x00 — Remote-control relay address(0x0000-0x0003)

Slave PC response: 0x01 0x05 0x00 0x00 0xFF 0x00 CRC

(4) Remote-control multi-loop switching output status (Function code 0 X 0F)

Host PC request: 0x01 0x0F 0x00 0x00 0x00 0x04 0x01 0x0F CRC

- 0x0F — Action value(0x0F 4 loops close)
- 0x01 — Data Byte numbers
- 0x00 0x04 — Loop quantity of remote-control relay (fixed 0 X 0004)
- 0x00 0x00 — Starting address of remote-control relay (Fixed 0 X 0000)
- 0x01 — Function code of remote-control multi-loop relay 0x0F

Slave PC response: 0x01 0x0F 0x000x00 0x00 0x04 CRC

The communication protocol of this meter follows the standard MODBUS-RTU protocol. And in the communication address Table, there are both floating-point primary power grid data and secondary power grid fixed-point integers. Customers can choose to read the corresponding data according to their own system requirements. MODBUS protocol communication address information table is shown in Table 17.

Communication message examples:
Read data(function code:03): this function makes the user get the data collected and recorded by terminal device and The system parameter. There is no quantity limit when the host PC collect the data in one time, but can't exceed the range of defined address. The below example is from terminal device addressing 12(0CH), read 3 data Ia, Ib, Ic (each address take 2 byte, the beginning address of Ia is 43 (2BH) , data length is 3(03H)byte.

Query the data frame (Host PC) (See Table 13)

Table 13

Address	Command	The beginning register address (high)	The beginning register address (low)	Register quantity (high)	Register quantity (low)	CRC16 (low)	CRC16 (High)
01H	03H	00H	58H	00H	02H	45H	D8H

Response data frame (slave PC) (see Table 14)
It means Ia=0000C317H(4.9943).

Table 14

Address	Command	Data length	Data 1234	CRC16 (low)	CRC16 (High)
01H	03H	04H	00H 00H C3H 17H	72H	E5H

Preset data (function code: 16): this function allows to change the contents of multiple registers (electrical measurement can be written with this function number, it should be emphasized that the written data is a writable attribute parameter, the number does not exceed the address Range), the following example is a communication method with a current transformation ratio of 400 / 5A = 80.

Query the data frame (Host PC) (See Table 15)

Table 15

Address	Command	The beginning register address (high)	The beginning register address (low)	Register quantity (high)	Register quantity (low)	Number of characters	Written data	CRC16 (low)	CRC16 (High)
01	10	00	2E	00	01	02	0050	A0	22

Query the data frame (Slave PC),indicates that the data has been written (See Table 16)

Table 16

Address	Command	The beginning register address (high)	The beginning register address (low)	Register quantity (high)	Register quantity (low)	CRC16 (low)	CRC16 (High)
01	10	00	2E	00	01	61	C0

MODBUS Address Data Table (see Table 17)

Table 17

Address		Description	Data format	Data length	Read/ write	Explanation
0001						
0002		Corresponding item of output 1	uchar	1	R/W	Corresponding item of switching output
		Level delay of output 1	uchar	1	R/W	Unit: Second
0003		Corresponding value of output 1	uint	2	R/W	Alarm value
0004		Return difference value of output 1	uint	2	R/W	Returned value
0005		Limit alarm of output 1	uint	2	R/W	Min. value of lower alarm
0006-0019		Output 2-6 loops				
001A		Corresponding item of analog output 1	uchar	1	R/W	Corresponding item of analog output
		Corresponding mode of analog output 1	uchar	1	R/W	0,4-20mA 1,0-20mA
001B		Corresponding value of analog output 1	uint	2	R/W	Analog corresponding value
001C-0021		2-4 loops of analog output				
0022-0027		Backup				
0028		Power display setting	uchar	1	R/W	Reserve
		Wiring mode selection	uchar	1	R/W	see the address explanation
0029		Power-on display	uchar	1	R/W	displayed contents
		Back-light delay time	uchar	1	R/W	Back-light delay time
002A		Luminance	uchar	1	R/W	LED Luminance
		CT quantity	uchar	1	R/W	0,3CT 1,2CT
002B		Time setting of scrolling display				
002C		Voltage multiplying power	uint	2	R/W	1-9999
002D		PT divisor	uint	2		1-9999
002E		Current multiplying	uint	2	R/W	1-9999
002F		Backup				
0030		Voltage null	uint	2	R/W	0.001-9.999
0031		Current null	uint	2	R/W	0.001-.500
0032		Power null	uint	2	R/W	0.001-9.999
0033		Communication protocol	uint	2	R/W	0-RS485Modbus
0034		Communication address	uint	2	R/W	0-247
0035		Baud rate	uint	2	R/W	1-6,1:1200,2:2400, 3:4800,4:9600(default), 5:19200,6:38400

0036		checking	uint	2	R/W	0-2,0:even checking, 1:odd checking, 2:no checking(default)
0037		Backup				
0038-003B		645 communication address	uchar	8	R/W	BCD code
003C		Baud rate of No.2 loop	uint	2	R/W	1-6,1:1200,2:2400, 3:4800,4:9600(default), 5:19200,6:38400
003D		No.2 loop checking	uint	2	R/W	0-2,0:even checking, 1:odd checking, 2:no checking(default)
003E-003F		Backup				
0040		Switching output information	uint	2	R	High 8 bit backup, low 8 bit valid 0: OFF 1: ON
0041		Switching input information	uint	2	R	0: OFF 1: ON
0042 0043		A phase voltage	ulong	4	R	4 decimal point (0.0001V)
0044 0045		B phase voltage	ulong	4	R	4 decimal point (0.0001V)
0046 0047		C phase voltage	ulong	4	R	4 decimal point (0.0001V)
0048 0049		Average phase voltage	ulong	4	R	4 decimal point (0.0001V)
004A 004B		Phase voltage unbalance	ulong	4	R	0.01-99.99 2 decimal point(%)
004C 004D		AB line voltage	ulong	4	R	4 decimal point (0.0001V)
004E 004F		BC line voltage	ulong	4	R	4 decimal point (0.0001V)
0050 0051		CA line voltage	ulong	4	R	4 decimal point (0.0001V)
0052 0053		Average line voltage	ulong	4	R	4 decimal point (0.0001V)
0054 0055		Zero-sequence voltage	ulong	4	R	4 decimal point (0.0001V)

0056 0057		Line voltage unbalance	ulong	4	R	0.01-99.99 2 decimal point (%)
0058 0059		A phase current	ulong	4	R	4 decimal point (0.0001A)
005A 005B		B phase current	ulong	4	R	4 decimal point (0.0001A)
005C 005D		C phase current	ulong	4	R	4 decimal point (0.0001A)
005E 005F		Average current	ulong	4	R	4 decimal point (0.0001A)
0060 0061		Zero-sequence current	ulong	4	R	4 decimal point (0.0001A)
0062 0063		Current unbalance	ulong	4	R	0.01-99.99 2 decimal point (%)
0064 0065		A phase active power	long	4	R	4 decimal point (0. 0001kW)
0066 0067		B phase active power	long	4	R	4 decimal point (0. 0001kW)
0068 0069		C phase active power	long	4	R	4 decimal point (0. 0001kW)
006A 006B		Total active power	long	4	R	4 decimal point (0. 0001kW)
006C 006D		A phase reactive power	long	4	R	4 decimal point (0. 0001kvar)
006E 006F		B phase reactive power	long	4	R	4 decimal point (0. 0001kvar)
0070 0071		C phase reactive power	long	4	R	4 decimal point (0. 0001kvar)
0072 0073		Total reactive power	long	4	R	4 decimal point (0. 0001kvar)
0074 0075		A phase apparent power	long	4	R	4 decimal point (0. 0001kVA)

0076 0077		B phase apparent power	long	4	R	4 decimal point (0. 0001kVA)
0078 0079		B phase apparent power	long	4	R	4 decimal point (0. 0001kVA)
007A 007B		Total apparent power	long	4	R	4 decimal point (0. 0001kVA)
007C		A phase power factor	int	2	R	3 decimal point
007D		B phase power factor	int	2	R	3 decimal point
007E		C phase power factor	int	2	R	3 decimal point
007F		Total power factor	int	2	R	3 decimal point
0080		Frequency	uint	2	R	Frequency 2 decimal point
0081		Backup				
0082 0083		Total positive active energy	ulong	3	R	Secondary side (0.001kWH)
0084 0085		Total negative active energy	ulong	3	R	Secondary side (0.001kWH)
0086 0087		Total positive reactive energy	ulong	3	R	Secondary side (0.001kvarH)
0088 0089		Total negative reactive energy	ulong	3	R	Secondary side (0.001kvarH)
008A 008B		Total positive active energy	float		R	Primary side
008C 008D		Total negative active energy	float		R	Primary side
008E 008F		Total positive reactive energy	float		R	Primary side
0090 0091		Total negative reactive energy	float		R	Primary side
0092-0093		Total Positive Apparent Energy	ulong	4	R	Secondary Value(0.001kVAh)
0094-0095		Total Negative Apparent Energy	ulong	4	R	Secondary Value(0.001kVAh)
0096-0097		Total Positive Apparent Energy	float	4	R	Primary Value
0098-0099		Total Negative Apparent Energy	float	4	R	Primary Value

8. Function output

8.1 Energy measurement and pulse output

This series meter adopt 3 row 12 digit to display primary energy, VH-0 show total active energy, VH-1 show negative active energy, VAHO show negative reactive energy, VAH1 show negative reactive energy, pulse normal value is 8000imp/kwh. (VH-0)=(VH-0)+(VH-1) VAH0=VAH0+VAH1

8.2 switching value part:

This series meter supply 2 loops of switching value input function and 2 loops of switching value output function , 4 loops of switching input adopt dry contact resistance switch signal input, The inside of meter is equipped with 12V working power supply and no need external power supply.When connect to external power supply, the switch input module D1 of meter will collect the POWER-ON information and the meter will display 1. when disconnect to external power,the switch input module D1 of meter will collect POWER-OFF information and the meter will display 0. Switch input module not only can collect and display switch message,but also can realize remote-control transmission function through RS485 interface, which is remote-communication function. 4 loops of optocoupler relay output (switching value output) is applied to many situation such as alarm indication, protection and control function.If the switch output is effective, and the relay is working, the meter will display 1. If the switch output is not effective, and the relay is closed, the meter will display 0.

8.2.1 Electrical parameter:

switch input DI: Power-on resistance R<5000;c Power-off resistance >100KΩ

switch out DO: AC 250V ,0.1A

8.2.2 Register (see Table 18)

DI0 information register(0X0021): this register indicate the status and information of 12 loops of switch input and 4 loops of switch output. Table 18

Di0 register	BIT15	BIT14	BIT13	BIT12	BIT11	BIT10	BIT9	BIT8	BIT7	BIT6	BIT5	BIT4	BIT3	BIT2	BIT1	BIT0
Corresponding switch port	DI1	DI2	DI3	DI4	DI5	DI6	DI7	DI8	DI9	DI10	DI11	DI12	DO1	DO2	DO3	DO4
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Di0 information register low 4 bit(BIT3,BIT2,BIT1,BIT0) is switch output status. Register display 000000000000 0101 means that switch output port 2,4 loop is power on;1,3 way is power off;DI0 message register high 12 bit(BIT15-BIT4) is switch input status. Register display 1101 000000000000 means that switch input port the 1,,2,4 loop is power on, other loops is power off. All DI0 information will display on the screen.The alarm output parameters of each switch are stored in

3 consecutive address spaces of DOSI. If the first loop use 3 bytes of address 10, 11,12(BTYE2,BYTE1,BYTE0) to save. Address lowest byte(address 10)save the parameter of alarm output.Ua low alarm parameter is 1,high alarm parameter is 129; 0 means remote mode. other 2 byte (address 11,12) is alarm over limit parameter. Other 3 loops is similar. Corresponding address refer to the below table.(See Table 19)

Table 19

Item	Variate	Meaning:DOSI (BYTE2,BYTE1,BYTE0)
Switch output 1	DO 1	BYTE2 (1~255),alarm item 1-26 corresponding to lower alarm of measured electric quantity items.129-154 corresponding to higher alarm. 0 indicate reserve method. the details refer to the table of switching value output, analog output energy parameter. BYTE 10(1~9999). alarm limit parameter,the data format is same to electric quantity format, the decimal point should be noticed.
Switch output 2	DO 2	
Switch output 3	DO 3	
Switch output 4	DO 4	

8.2.3 Application example:

A. switch input function :

Switch module is provided with 12 loops of switch input collect function. After collecting the input signal, the meter display "ON 1" or "OFF 0 "information to monitor switch signal. Switchover the meter to switch message display status, "DI" indication light will ON. refer to Page 20 DISP=13 Fig.

Through meter RS485 protocol, the message of switch register(DI 0) can be transmitted to remote PC terminal.

B. switch output function:

It can control 2 loops of switch output through upper computer writing in control message to DI 0 message register. Write in 1 corresponding Port ON, write in 0 corresponding port OFF;for example: If write in binary number 1 0 0 0 0000, it indicate 1 loop output port is ON, 2 loop is OFF. This function can't be used with another limit alarm function of switch output module in the same time. if using remote function, need set the power parameter to 0, close alarm output function, and set the second line of switching value setting to 0. please refer to Page 21 DISP=14. under remote-control status, No.1 loop is ON status, No.2 is OFF status.

Another function of switch output module is over limit alarm output. when the electric parameter over the set range, the corresponding switch output port is ON status, the meter show 1 on the corresponding position, when signal return back to the range, the meter show 0.

4th loop is A phase voltage :TYPE is 129,UAL is 3800;3800 corresponding to secondary side voltage 380V.
Total active power: TYPE is 141, UAL is 3300;3300 corresponding to secondary power value 3300W.
Total power factor: TYPE is 149,UAL is 1000;1000 corresponding to secondary power factor value 1.000.
Frequency: TYPE is 154,UAL is 5000;5000 corresponding to secondary frequency value 50.00Hz
Notice: when TYPE set 0000,mean " remote-control " status.

9.Ordinary problem and solution
9.1 About communication
● Meter no data return back
Ans:
First make ensure the setting of mater communication such as slave address, baud rate, checkout mode meet the requirements of master PC.
If there is many meters don't send back the data, detect if the connection of communication bus is correct and reliable and Rs485 converter is normal.If only one or few meter communication is abnormal ,also need check corresponding communication wires.the user can revise the address of abnormal and normal meter, remove or confirm the master software problem , or exchange the installation position of abnormal and normal meter installation position, to exclude or confirm the meter fault.
● The data returned by meter not correct
Ans:
This series meters communication data for customer is primary grid float type and secondary grid int/long type. Please read carefully about data store address and data store format in communication address table, and make ensure converting according to corresponding data format.
9.2 About U, I , P measurement not accurate
Ans:
First make sure the correct voltage and current signal have already connected to the meter. The voltage signal can be measured by the meter, the current signal can be measured by clamp meter. Second, make sure the signal wiring is correct. for example. the input terminal of current signal, and the phase sequence. This meter have power display , only under reverse current delivery status, the active power is negative. under normal status.active power symbol is positive. if active

power symbol is negative , the current input and output wires maybe connect wrong, or maybe phase sequence is incorrect. And the displayed energy is primary grid value, if the multiple rate setting of voltage/current transformer is different to actual rate,it will result in inaccurate display of energy. The measurement range can't change after leaving factory.
The wiring can be revised according to on-site practical situation, Connect, but the wiring method setting in programming menu should be same to the actual method, otherwise will result in error display.

9.3 About the energy counting not accurate

Ans

The energy accumulation is based on the power, first check if the meter power value conform to practice load. This series meter can support two-way energy accumulation. Under incorrect wiring situation, total active power is negative, the energy accumulation to negative active power, negative active power can't accumulate. The normal problem happened in on-site is current transformer input and output wiring reversed. This series product can check the active power symbol of each phase,if power is negative, the wring maybe wrong. Phase sequence wiring wrong will result in energy counting not accurate.

9.4 Meter no bright

Ans

Make ensure connect correct auxiliary power(refer to the label on the cover) already connect to the auxiliary power terminal of meter. If the auxiliary power over the range,the meter will be damaged and can't recover. The meter can be used to measure the voltage of auxiliary power, if power voltage is normal, meter still not have any display, please power-off then power-on again, If meter still display abnormally, please contact with our technical dept.

9.5 Meter no respond to any operation

Ans:

Press meter button " SET " " ← " " → " " ↵ " to power-off the meter then power-on again. If the meter can't back to normal, please contact our technical dept.

9.6 Other abnormal condition

Ans:

Please contact our technical service department , the user should describe the details of the situation, our technical staff will analyse the possible reasons according to your feedback. If the problem that cannot be resolved after communication, we will arrange technical personal to on-site to solve the problem.